Green Research Software Engineering for Beginners

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Why Green Software Matters

World bank in [1] says that ICT sector's contribution to global GHG emissions is atleast 1.7%. Scoping the emissions helps identify where the emissions are coming from for better carbon accounting.

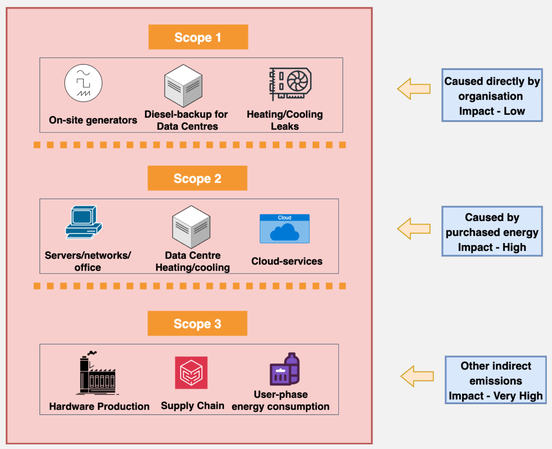


Figure: Scopes of emissions (GHG protocol)

Energy-intensive software processes

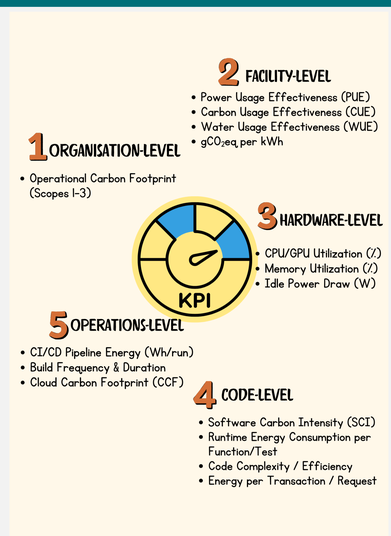
- Large-Scale Simulations & Modeling [2]
- Machine Learning & Deep Learning [3]
- High-Resolution Data Processing [4]
- Computational Fluid Dynamics (CFD) & Finite Element Analysis (FEA) [5]
- Monte Carlo Simulations [6]
- Big Data Analytics [7]
- Data Storage & Transfer [8]
- Inefficient Software & Poorly Optimized Code [9]

Core Principles

The GREENER principles [10] could help Researchers and Research Software Engineers to build their road to Greener Software by following the GREENER framework.



Energy & Carbon Tracking



Call to action for RSEs

Green Software Engineering is a work in progress where stakeholders are still identifying the leaks and inefficiencies. But there is a holistic step-by-step guide that RSEs can begin with.



Figure: Guide to go green

Tools and Technologies

Libraries

- *CodeCarbon*: python code emissions
- *Eco2AI*: AI/ML model emissions

Software Tools

- *Green Algorithms*: emissions of computational tasks [11]
- *Green Metrics Tool*: automated SCI measurements [12]

Training Material

- *Green Software* [13]

References

[1] World Bank.
Measuring the emissions and energy footprint of the ict sector: Implications for climate action, 2023.
Accessed: 2025-03-28.

[2] Jack Dongarra, Piotr Luszczek, and Thomas Sterling.
The importance of being low power in high performance computing.
Cyberinfrastructure Technology Watch, 3(3), 2007.
Accessed: 2025-03-28.

[3] World Economic Forum.
Ai and energy: Will ai help reduce emissions or increase demand?
2024.
Accessed: 2025-03-28.

[4] Author(s) Unknown.
A review on the decarbonization of high-performance computing: Current trends and future directions.
Renewable and Sustainable Energy Reviews, Unspecified(Unspecified):Unspecified, 2023.
Accessed: 2025-03-28.

[5] PRACE.
Electricity in hpc centres: A white paper.
Technical report, Partnership for Advanced Computing in Europe (PRACE), 2021.
Accessed: 2025-03-28.

[6] Luis G. León-Vega, Niccolò Tosato, and Stefano Cozzini.
A comprehensive analysis of process energy consumption on multi-socket systems with gpus, 2024.

[7] Frontier Group.
Fact file: Computing is using more energy than ever, 2024.
Accessed: 2025-03-28.

[8] Granulate.
Data center energy consumption: Drivers, metrics optimization, 2024.
Accessed: 2025-03-28.

[9] Sung Une Lee, Niroshinie Fernando, Kevin Lee, and Jean-Guy Schneider.
A survey of energy concerns for software engineering.
Journal of Systems and Software, 210:111944, 2024.

[10] A. Greener and Others.
Towards greener algorithms: A framework for energy-aware computing.
Nature Computational Science, 2023.
Accessed: 2025-03-28.

[11] Green Algorithms Project.
Green algorithms - estimating the carbon footprint of computation, 2024.
Accessed: 2025-03-28.

[12] Green Coding Solutions GmbH.
Green metrics tool, 2025.
Accessed: 2025-09-08.

[13] Green Software Foundation.
Introduction to green software engineering, 2024.
Accessed: 2025-03-28.

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